AUTOMATED QUESTION PAPER

PARSING & PATTERN MINING

PYQ

Analyser

Evidence-Based Exam Intelligence

COURSE

BCSE307L — Compiler Design

GENERATED

26 March 2026

REPORT SECTIONS

01 Topic Importance Rankings

02 Frequency Analysis

03 Repeated Questions Detected

04 Grading Distribution

05 Question Style Analysis

MODULE 2

1 Parser 10× · 31.0 marks

2 CLR Parsing - LALR Parsing 8× · 26.0 marks

MODULE 3

1 Syntax Directed Translation Schemes 4× · 15.3 marks

2 Syntax Directed Definition 4× · 6.8 marks

MODULE 4

1 Three Address Code 13× · 55.0 marks

2 Control Flow 1× · 10.0 marks

MODULE 2

Parser	10
CLR Parsing - LALR Parsing	8

MODULE 3

Syntax Directed Translation Schemes	4
Syntax Directed Definition	4

MODULE 4

Three Address Code	13
Control Flow	1

PARSER

> *List the canonical collection of sets of LR(1) items for the given grammar*

Asked 6 times

cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 1_i) | cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 1_ii)(a) | cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 1_ii)(b) | cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 2_i) | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 1_b | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 1_d

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> *for the grammar.(2 M) Show the moves of the parser for the input string a + a (a + aj#*

Asked 2 times

cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 1_iii) | cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 2_iii)

CLR PARSING - LALR PARSING

> *Construct SLR parsing table for the grammar.(3 M)*

Asked 6 times

cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 1_ii)(b) | cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 2_ii) | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 1_a | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 1_c | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 1_d | cleaned_3. BCSE307L-CAT-2-A2-2023-2024.json - 1_a

SYNTAX DIRECTED TRANSLATION SCHEMES

> *Construct a syntax directed translation scheme that translates arithmetic expressions from postfix notation into infix notation. Give annotated parse tree for the expressions: 95-2* and 952*-.*

Asked 2 times

cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 3 | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 2_a

THREE ADDRESS CODE

> *Generate the three address code for the following statements using quadruples: a) x=(atb)*(c-d)/e+f*(g-h)/(i+j)-k*I/m b) if(n>10 && n<30) p=p +l*

Asked 9 times

cleaned_1. BCSE307L-CAT-2-C1-2023-2024.json - 4 | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 2_b | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 3_a | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 3_b | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 3_c | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 5_a | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 5_b | cleaned_3. BCSE307L-CAT-2-A2-2023-2024.json - 3 | cleaned_3. BCSE307L-CAT-2-A2-2023-2024.json - 4

> *Answer the followings and design a syntax-directed definition for calculating the storage layout of a three-dimensional array declared with zero-based indexing. Consider the array declaration format: T A[d1][d2][d3]; Where: T is a primitive data type (int, float, or char), d1, d2, and d3 are the dimension sizes. a) Complete the missing part of the grammar rule to represent the array declaration: D — T ID [NUM 1] [NUM2] [NUM3] ; T — int | float | char*

Asked 2 times

cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 4_a | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 4_b

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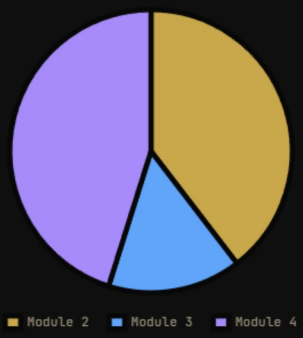
> *Basic Blocks (3 Marks)*

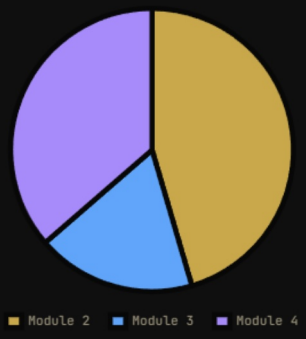
Asked 2 times

cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 5_c | cleaned_2. BCSE307L-CAT-2-A1-2025-2026.json - 5_d

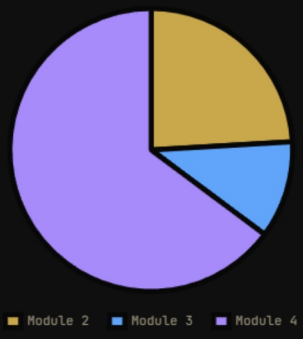
COMBINED AVERAGE — ALL PAPERS



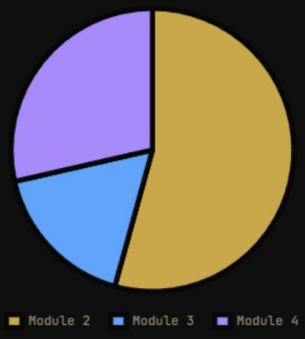
1. BCSE307L-CAT-2-C1-2023-2024



2. BCSE307L-CAT-2-A1-2025-2026



3. BCSE307L-CAT-2-A2-2023-2024



NUMERICAL

66.7%

Calculation-heavy questions. Expect formulae, step-by-step derivations, and numerical workings.

DESCRIPTIVE

33.3%

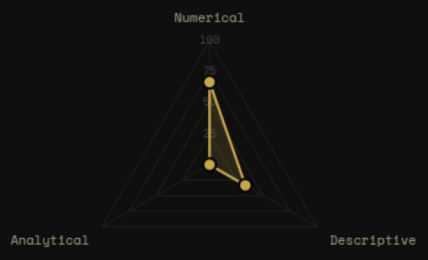
Theory and explanation-based questions. Focus on conceptual clarity and structured answers.

ANALYTICAL

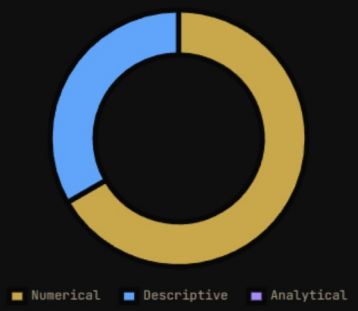
0.0%

Case studies and reasoning questions. Apply concepts to novel scenarios.

STYLE PROFILE — COMBINED RADAR



COMBINED DISTRIBUTION — DOUGHNUT



PER-PAPER STYLE BREAKDOWN

1. BCSE307L-CAT-2-C1-2023-2024 — Numerical: 60.0% · Descriptive: 40.0% · Analytical: 0.0%

2. BCSE307L-CAT-2-A1-2025-2026 — Numerical: 70.6% · Descriptive: 29.4% · Analytical: 0.0%

3. BCSE307L-CAT-2-A2-2023-2024 — Numerical: 66.7% · Descriptive: 33.3% · Analytical: 0.0%

Numerical Descriptive Analytical

DOMINANT STYLE

Numerical questions make up 66.7% of all papers. This should define your primary preparation approach.

PAPERS ANALYSED

Style patterns computed from 3 question papers. Higher the paper count, more reliable the distribution signal.

LEAST ASKED

Analytical questions are least frequent at 0.0%. Do not skip — even low-frequency types carry marks.

REPORT COMPLETE

All insights derived solely from uploaded question papers.

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